SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

(Deemed to be University) • Chennai

ECA1804 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO4 INVESTIGATIVE ASSIGNMENT

Ultra-High-Speed Photonic Transport Link: Technology Evaluation, Impairment Mitigation, and Integrated System Design

32-channel DWDM, 200 Gb/s per channel (6.4 Tb/s) – DP-16QAM, Silicon Photonics, Hybrid Raman-EDFA and Coherent DSP • (1 × 100 = 100 Marks)

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Submission Date	24 October 2026

Report workflow: REQUIREMENT → LITERATURE → ALTERNATIVES → ANALYSIS → SELECTION → SYSTEM DESIGN → VERIFICATION → FINAL DECISION

Executive summary

This report investigates a 32-channel, 200 Gb/s-per-channel DWDM transport link (6.4 Tb/s aggregate) and defends an end-to-end design. After comparing the candidate technologies quantitatively, the selected architecture is: **dual-polarisation 16-QAM at 28.75 GBd** on the 50 GHz ITU grid, generated by a **silicon-photonic dual-polarisation IQ Mach-Zehnder modulator**; **hybrid backward-Raman + EDFA** amplification of 8 spans of 80 km SMF-28 (640 km) with gain-flattening filters; and a **coherent receiver with a digital signal-processing chain** (frequency-domain CD equalisation, 2×2 MIMO polarisation demultiplexing, carrier-phase recovery and soft-decision FEC), with single-channel digital back-propagation kept as an optional upgrade.

Item	Result of the investigation
Aggregate capacity	6.4 Tb/s net (32 × 200 Gb/s); 7.36 Tb/s line rate with 15 % FEC; 1.6 THz of the ~4.4 THz C-band
Spectral efficiency	4.0 b/s/Hz net (4.6 b/s/Hz gross) - 1.3× the efficiency of DP-QPSK, 3× that of PAM4
Link OSNR after 8 spans	29.4 dB (0.1 nm) against 16.3 dB required (+1.5 dB implementation = 17.8 dB)
Q-factor at design point	8.67 dB analytical (GN model), 8.22 dB simulated; FEC threshold 6.25 dB (BER 2×10^{-2})
Margin over FEC threshold	2.6 dB (analytical) / 2.1 dB (split-step simulation) in SNR
Decision	**YES - deploy** (up to 8 × 80 km), with the conditions listed in the Final Engineering Decision

Assumed data and method (please read)

The assignment text leaves several values blank. They were fixed as below so that every calculation is reproducible; change them in the equations if your faculty supplies different values.

Parameter	Value used	Reason
Amplifier input / target output per channel	-20 dBm / 0 dBm	Blank in the question; typical for an 80 km span with 20 dB loss and a 0 dBm launch
Span	80 km SMF-28, 0.2 dB/km + 4 dB connectors, splices, margin = 20 dB	Standard terrestrial span
Fibre	D = 17 ps/(nm km), S = 0.056, $\gamma = 1.3$ $W^{-1}km^{-1}$, $A_{eff} = 80$ µm², PMD = 0.1 ps/√km	SMF-28 data-sheet class values (blank in the question)
Grid, FEC	50 GHz (ITU-T G.694.1), 15 % soft-decision FEC, pre-FEC BER limit 2×10^{-2}	Typical SD-FEC; threshold depends on the code chosen
Transceiver back-to-back SNR	16.5 dB	Typical integrated 16-QAM transceiver (DAC/ADC, driver, receiver noise)
Distance	8 spans × 80 km = 640 km	Chosen by the designer; reach vs distance is analysed in Stage 5

About the simulations. OptiSystem / OptSim were not available in this environment, so the simulations in this report were written from scratch in Python (NumPy) - a full split-step Fourier solution of the dual-polarisation Manakov equation for 3 WDM channels, ASE noise injection, a 2×2 MIMO equaliser and BER counting - which is the same model those tools solve. Fig. D gives the component-by-component OptiSystem layout so the design can be rebuilt there; the numbers in the tables are from the Python runs and the analytical equations. Equipment power, footprint and device values quoted from the literature or industry practice are typical figures and are marked with '~' or 'approx.'.

STAGE 1 - INTERPRET THE COMMUNICATION REQUIREMENT

15 Marks | K4 - Analyze

1.1 Literature and standards review

Source	Relevance to this design
ITU-T G.694.1 (DWDM frequency grid); ITU-T G.698.2 (amplified multichannel DWDM interfaces)	Defines the 12.5 / 25 / 50 / 100 GHz anchored-at-193.1 THz grid; 50 GHz chosen. G.698.x give the multichannel optical-interface framework (single-channel and amplified links).
IEEE 802.3ct (100 Gb/s over DWDM, coherent) and IEEE 802.3ck (100/200/400 Gb/s electrical interfaces, 106.25 GBd PAM4); OIF-CEI-112G	802.3ck / CEI-112G define the host (electrical) side that feeds the line card; 802.3ct is the nearest IEEE coherent-DWDM line-side PHY. OIF 400ZR (DP-16QAM, ~60 GBd, 75 GHz) is the closest DP-16QAM industry reference.
Essiambre et al., J. Lightwave Technol. 28(4), 2010	Nonlinear Shannon limit: capacity of fibre is limited by Kerr nonlinearity - sets the optimum launch power and the SNR ceiling.
Kikuchi, J. Lightwave Technol. 34(1), 2016; Ip et al., Opt. Express 16(2), 2008	Principles of coherent detection, DSP receiver architecture.
Savory, Opt. Express 16(2), 2008	Digital filters for CD compensation and polarisation demultiplexing; tap-count rule used in Stage 4.
Poggiolini, J. Lightwave Technol. 30(24), 2012	GN model of nonlinear interference - used for the analytical Q-factor vs launch-power curves.
Bromage, J. Lightwave Technol. 22(1), 2004	Raman amplification principles, effective noise figure, double-Rayleigh scattering.
Reed et al., Nature Photonics 4, 2010; Wang et al., Nature 562, 2018	Silicon and thin-film lithium-niobate modulators: bandwidth, $V_{\pi}L$, integration.
Ip & Kahn, J. Lightwave Technol. 26(20), 2008	Digital back-propagation: gain vs complexity.

1.2 Aggregate capacity and spectral budget

Aggregate capacity = 32 channels × 200 Gb/s = **6.4 Tb/s** (net, client side). With 15 % FEC overhead each wavelength carries a line rate of 200 × 1.15 = **230 Gb/s** (7.36 Tb/s in total). The 32 carriers on a 50 GHz grid occupy 32 × 50 GHz = **1.6 THz** (from 192.325 to 193.875 THz, i.e. 1546.3 - 1558.8 nm), about 36 % of the 4.4 THz C-band, so the design leaves room to grow to 88 channels. **Net spectral efficiency** = 200 Gb/s / 50 GHz = **4.0 b/s/Hz** per channel (6.4 Tb/s / 1.6 THz); the total, including FEC, is 4.6 b/s/Hz.

1.3 Physical-layer metrics to be investigated

Metric	Definition / target	Why it matters
Spectral efficiency	SE = R_{net}/channel spacing = 4 b/s/Hz	Determines how many channels fit in the C-band
OSNR (0.1 nm)	OSNR = $SNR_{sym} \cdot R_s / B_{ref}$, B_{ref} = 12.5 GHz	Amplifier noise budget; sets reach
BER / Q-factor	Pre-FEC BER ≤ 2×10^{-2}; Q = $\sqrt{2}\,\mathrm{erfc}^{-1}(2\,\mathrm{BER})$ = 2.05 (6.25 dB)	Defines the working limit of the SD-FEC
Launch power per channel	0 dBm (+15 dBm total) near the nonlinear optimum	Balance between ASE noise and Kerr nonlinearity
Modulation bandwidth	≥ 0.7 R_s = 20 GHz	Transmitter component selection
Insertion loss	Tx PIC, AWG mux/demux, span loss 20 dB	Power budget, OSNR at transmitter
Thermal / footprint	Si PIC chip, CFP2-class form factor, thermal tuning power	Integration scalability for 32 wavelengths

1.4 Impairments and the DSP function that handles each

Impairment	Physical origin	Effect on the signal	Matching compensation
Chromatic dispersion (CD)	Wavelength-dependent group velocity, D = 17 ps/(nm km)	Accumulated 10,880 ps/nm over 640 km spreads each symbol over ~72 symbols	Static frequency-domain / FIR equaliser
PMD	Random birefringence, mean DGD = 0.1√640 = 2.5 ps	Polarisation mixing, ISI (small: 0.07 symbol)	Adaptive 2×2 MIMO equaliser (CMA/RDE)
SPM / XPM / FWM	Kerr effect, $n_2 I$	Intensity-dependent phase noise, inter-channel crosstalk (NLI)	Optimum launch power; DBP (partial)
ASE noise	EDFA / Raman spontaneous emission	SNR degradation, irreducible by DSP	Low-noise (hybrid) amplification
Laser phase noise, frequency offset	Tx and LO linewidth, detuning	Constellation rotation	Frequency-offset + carrier-phase estimation (BPS)
Filter crosstalk, PDL, IQ imbalance	AWG/MMI leakage, component PDL, receiver skew	Linear crosstalk, SNR loss	Front-end orthogonalisation, margin allocation

1.5 Practical trade-offs: Silicon photonics vs lithium niobate vs polymer

Platform	Strength	Weakness	Verdict for 32 wavelengths
Silicon photonics (SiP)	CMOS foundry, thousands of devices per wafer, co-packaged with Ge photodiodes and drivers, small footprint	Plasma-dispersion modulator: high loss (~6 dB), $V_\pi L$ ~ 1.5 - 2.5 V cm, chirp-free only in push-pull; ~35 - 45 GHz bandwidth	Best for volume and integration
Thin-film $LiNbO_3$	Linear, low loss, V_π ~ 3 V (calculated below), > 70 GHz	Costlier, less mature foundry, larger chip	Best performance; use if > 64 GBd is needed
EO polymer / plasmonic hybrid	Very high r_{33}, $V_\pi L$ < 1 V cm, compact	Thermal / photochemical ageing, reliability for 25 years not proven	Promising, higher deployment risk

Investigation Requirement Statement

The DWDM system shall carry 32 channels on the ITU-T 50 GHz grid (C-band, 192.325 - 193.875 THz), each at 200 Gb/s net (230 Gb/s with 15 % SD-FEC), giving 6.4 Tb/s and a net spectral efficiency of 4 b/s/Hz. Over a 640 km amplified SMF-28 link (8 × 80 km, 20 dB spans) the received OSNR shall exceed 17.8 dB (0.1 nm), including 1.5 dB implementation penalty, with a pre-FEC BER below 2×10^{-2} (Q ≥ 6.25 dB) and at least 1.5 dB end-of-life margin. The transmitter shall be a compact integrated photonic circuit with ≥ 20 GHz modulation bandwidth; CD, PMD and carrier impairments shall be compensated in the coherent receiver DSP; the design shall interface with IEEE 802.3ck / OIF-CEI electrical interfaces and ITU-T G.694.1/G.698 optical grids.

STAGE 2 - INVESTIGATE AND SELECT THE MODULATION / MODULATOR CONFIGURATION

20 Marks | K4/K5 - Analyze and Evaluate

2.1 Required baud rate and channel spacing

For a net rate R_{net} = 200 Gb/s and 15 % FEC overhead the line rate is R_b = 230 Gb/s. The symbol rate is $R_s = R_b / (N_{pol} \cdot \log_2 M)$, and the occupied optical bandwidth $\approx R_s(1 + \beta)$, with a root-raised-cosine roll-off $\beta = 0.1$ for coherent formats.

Format	Bits / symbol	Baud rate R_s	Occupied bandwidth	Minimum channel spacing
DP-16QAM (coherent)	2 × 4 = 8	230 / 8 = **28.75 GBd**	31.6 GHz	**37.5 GHz (50 GHz used)**
DP-QPSK (coherent)	2 × 2 = 4	230 / 4 = 57.5 GBd	63.2 GHz	75 GHz
DP-64QAM (coherent)	2 × 6 = 12	230 / 12 = 19.17 GBd	21.1 GHz	25 GHz
PAM4 (IM-DD, 1 pol.)	2	230 / 2 = **115 GBd**	≈ 1.3 × 115 = 150 GHz	**150 GHz** (32 ch = 4.8 THz > C-band)
NRZ-OOK	1	230 GBd	≈ 2 × 230 = 460 GHz	> 460 GHz - impossible
DP-DQPSK (differential)	4	57.5 GBd	63 GHz	75 GHz

Result: DP-16QAM needs only 28.75 GBd, so one wavelength fits inside the 50 GHz slot and electronics and modulators of the 25 - 35 GHz class are enough. PAM4 would need 115 GBd (4× faster), a 150 GHz slot, and 32 channels would not fit in the C-band at all; it also suffers severely from chromatic dispersion in the 1550 nm window (the 115 GBd spectrum spans ~1 nm and spreads by 10 ns/ps-class amounts after only tens of km).

2.2 Receiver sensitivity and required OSNR

For square Gray-coded M-QAM in AWGN, BER $\approx (4/\log_2 M)(1 - 1/\sqrt{M})\, Q(\sqrt{(3 \cdot SNR_s/(M-1))})$. Solving for BER = 2×10^{-2} gives the required symbol SNR; the OSNR in 0.1 nm (12.5 GHz) of a dual-polarisation signal is $OSNR = SNR_s \cdot R_s / B_{ref}$.

Format	Required SNR_s (dB)	R_s (GBd)	Required OSNR (dB, 0.1 nm)	SE net (b/s/Hz, 50/75/25 GHz slot)
DP-QPSK	6.25	57.5	12.88	200/75 = 2.67
DP-16QAM	**12.71**	28.75	**16.33**	**200/50 = 4.00**
DP-64QAM	18.43	19.17	20.29	200/37.5 = 5.33 (50 GHz grid wastes it)

Doubling the bits per symbol from QPSK to 16-QAM costs 3.4 dB OSNR (at constant net rate); 64-QAM costs another 4.0 dB and cuts the reach roughly in half again. DP-16QAM is therefore the 'knee': it already reaches 4 b/s/Hz and its required OSNR of 16.3 dB is easily met by an amplified 640 km link (29.4 dB achieved, Stage 3).

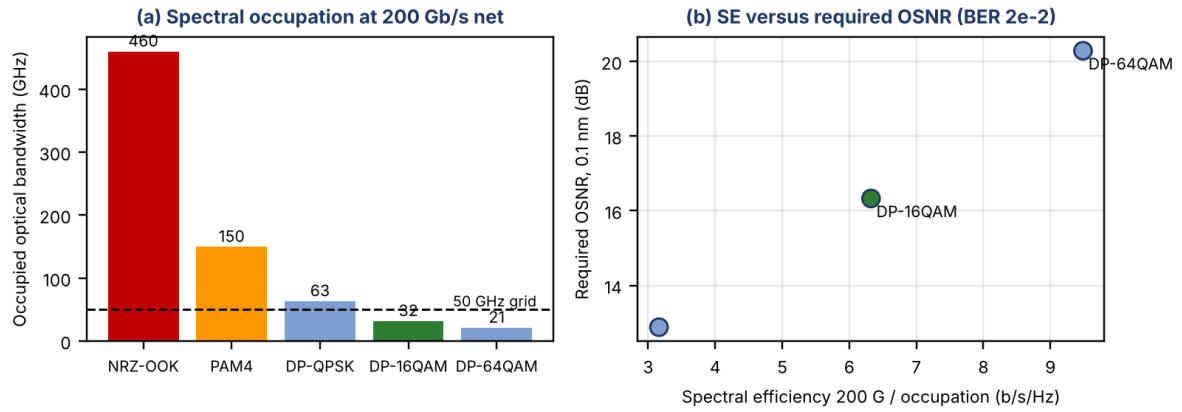


Fig. 2.1: (a) spectral occupation of the candidate formats at 200 Gb/s net; (b) spectral efficiency versus required OSNR for the coherent formats

2.3 Comparison of the candidate formats

Format	Spectral efficiency	Required OSNR / SNR	Receiver sensitivity	Tolerance to impairments	Complexity
NRZ-OOK	< 0.5 b/s/Hz at 200G	High (needs 230 GBd)	Direct detection, ~ -20 dBm class at 10 G only	Very poor CD / PMD	Lowest, but 230 GBd electronics do not exist
PAM4	1.3 b/s/Hz	Higher: 3 eye openings, ≈ -9.5 dB electrical SNR vs OOK	Direct detection, no phase information	CD limited to a few km at 115 GBd without DSP	Moderate, 115 GBd DAC/ADC
DPSK	< 1 b/s/Hz	Low (3 dB better than OOK)	Balanced delay-line interferometer	Moderate	Needs 230 GBd for 200G
DQPSK / DP-DQPSK	2.7 b/s/Hz	≈ 2 dB above coherent QPSK (14.9 dB)	Differential detection, no DSP gain	CD/PMD must be optically compensated	Interferometers, tight thermal control
16-QAM (single pol.)	2 b/s/Hz	As DP-16QAM, 3 dB less power per pol.	Coherent	DSP	115 GBd needed
DP-16QAM (coherent)	**4 b/s/Hz**	**16.3 dB**	**Coherent, shot-noise limited, LO gain**	**Full electronic CD / PMD compensation**	**ADC + DSP ASIC, 28.75 GBd**

2.4 Modulation mechanisms

The relevant requirement is: modulate the optical field with bandwidth > 20 GHz, low V_π, low chirp and low loss.

Mechanism	Physical principle	Speed limit / calculation	V_π / drive	Chirp / comment
Electro-optic (Pockels, $LiNbO_3$)	$\Delta n = \frac{1}{2} n^3 r E$ (linear)	Few fs intrinsic; electrode RF limited, > 70 GHz	$V_\pi = \lambda d / (n^3 r \Gamma L) = 5.89$ V (phase mod.), 2.94 V (push-pull MZM) for d = 8 µm, L = 1 cm, r_{33} = 30.8 pm/V	Zero chirp in push-pull

Mechanism	Physical principle	Speed limit / calculation	V_π / drive	Chirp / comment
Electro-optic (plasma dispersion, Si)	Carrier depletion in pn junction changes n and absorption	RC and transit time: ~35 - 45 GHz	$V_\pi L$ = 1.8 V cm; L = 4 mm → V_π = 4.5 V	Some residual chirp / amplitude-phase coupling
Acousto-optic	Photo-elastic grating from an acoustic wave	Acoustic transit time τ = D/v = 100 µm / 3.6 km/s = 27.8 ns → BW ≈ 0.35/τ = 12.6 MHz	RF power, not voltage	Far too slow for 28 GBd; used for tunable filters / frequency shifters
Thermo-optic	$dn/dT = 1.86\times10^{-4}$ K^{-1} (Si)	Thermal time constant 1 µs - 1 ms; ΔT for π in 500 µm = λ/(2L dn/dT) = 8.3 K	10 - 25 mW heater per π	Used only for bias / tuning / calibration
Magneto-optic (Faraday)	Polarisation rotation θ = V B L	Coil inductance → kHz - MHz	Magnetic field, not voltage	Non-reciprocal: used for isolators and circulators, not data modulators
Kerr effect ($\chi^{(3)}$)	$\Delta n = n_2 I$	fs response, but π shift over 1 m of silica needs I = λ/(2L n_2) = 3.0 GW/cm²	Optical pump, not electrical	Basis of SPM / XPM / FWM impairments, not practical for modulation

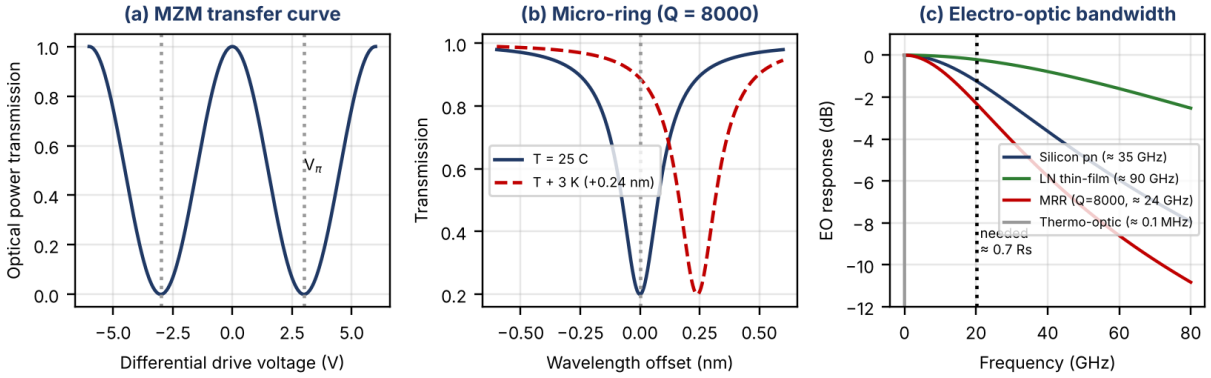


Fig. 2.2: (a) Mach-Zehnder transfer curve; (b) micro-ring resonance and its thermal shift; (c) electro-optic bandwidth of the candidates against the 0.7 R_s requirement

2.5 Mach-Zehnder (MZI) versus micro-ring (MRR) modulators

MZI. Output power $P = P_0 \cos^2(\pi V/2V_\pi)$ (Fig. 2.2a). Push-pull driving gives zero chirp and an amplitude-linear field transfer $\sin(\pi V/2V_\pi)$, which is exactly what a 4-level/16-QAM constellation needs. The balanced MZI is almost wavelength-insensitive (a 1 nm laser drift changes its phase only slightly) and its thermal drift is a bias phase only, removed by a slow control loop.

MRR. A ring of radius 10 µm and group index 4.2 has FSR = $\lambda^2/(n_g 2\pi R)$ = 9.1 nm. Its optical bandwidth is limited by photon lifetime: Δν = c/(λQ) = 48 / 24 / 10 GHz for Q = 4000 / 8000 / 20000. A high-Q ring (low V_π, low power) is therefore too narrow for 28.75 GBd, and a low-Q ring loses its efficiency. Thermal sensitivity is the decisive drawback: dλ/dT ≈ 0.08 nm/K against a linewidth of λ/Q = 0.19 nm, so **a 1 K change detunes the ring by 41 % of its linewidth** (Fig. 2.2b) and every ring needs its own thermal lock. Intensity-only rings also cannot produce a clean IQ (phase + amplitude) field without a ring-assisted MZI.

Parameter	SiP MZM (carrier depletion)	SiP micro-ring	EO-polymer MZM	Thin-film LN MZM
Electro-optic bandwidth	35 - 45 GHz	~ 24 GHz at Q = 8000 (photon-lifetime limit)	> 100 GHz demonstrated	> 70 GHz
Footprint	3 - 4 mm × 0.5 mm per MZM	~ 20 × 20 µm (≈ 1000× smaller)	1 - 5 mm	5 - 20 mm
V_π (calc.)	4.5 V (4 mm)	< 2 V (resonant enhancement)	~ 2 V ($V_\pi L$ ~ 1 V cm, 5 mm)	2.9 V
Insertion loss	~ 6 - 8 dB (incl. modulation)	< 1 dB off-resonance, ~ 3 dB on-resonance	~ 5 - 8 dB	~ 2 - 4 dB
Thermal sensitivity	Low: slow bias control	**Very high: 0.08 nm/K vs 0.19 nm line**	Low	Low (but DC drift)
Linearity / IQ capability	Good (nested MZM, 16-QAM demonstrated)	Poor for IQ	Good	Excellent
Integration with Ge PD, driver, AWG	**Excellent (CMOS foundry)**	Excellent	Hybrid process	Hybrid process

DESIGN DECISION 1 - SELECT THE PROPOSED MODULATION / MODULATOR CONFIGURATION

I recommend dual-polarisation 16-QAM at 28.75 GBd (230 Gb/s line rate) on the 50 GHz grid, generated by a silicon-photonic dual-polarisation IQ Mach-Zehnder modulator (four carrier-depletion pn travelling-wave MZMs, $V_\pi \approx 4.5$ V, 35 GHz bandwidth, 4 mm long) with thermo-optic bias control and, as the upgrade path, thin-film $LiNbO_3$ because it needs only 28.75 GBd (PAM4 needs 115 GBd and 150 GHz), reaches 4 b/s/Hz so all 32 channels use 1.6 THz, its required OSNR of 16.3 dB is 3.5 dB above QPSK but is met with 13.1 dB spare, and coherent detection lets the receiver DSP remove CD and PMD. The 35 GHz bandwidth exceeds the $0.7\ R_s = 20$ GHz requirement; the MZM is chirp-free, thermally robust and CMOS-integrable, whereas micro-rings (0.08 nm/K drift vs 0.19 nm linewidth) are unsuitable for 16-QAM IQ. Its main weakness (6 - 8 dB loss and $V_\pi \approx 4.5$ V, driven at ~3 V_{pp} per arm) is compensated by the booster amplifier and a SiGe driver.

STAGE 3 - DEVELOP AND EVALUATE THE OPTICAL AMPLIFICATION CONFIGURATION

20 Marks | K4/K5 - Analyze and Evaluate

3.1 Comparison of the amplifier technologies

Amplifier	Gain bandwidth	Saturated output power	Noise figure	Gain tilt / flatness	Inter-channel crosstalk	Remarks
EDFA	~ 35 nm (C-band)	+17 to +23 dBm	4.5 - 6 dB	Ripple ~ 5 dB, flattened with GFF	Negligible (slow gain dynamics)	Standard workhorse; lumped
SOA	30 - 100 nm	+10 to +15 dBm	7 - 10 dB	Polarisation dependent, nonlinear	**Strong (fast gain dynamics: XGM, FWM)**	Compact, switchable; unsuitable for 32-channel WDM line amplifier
Distributed Raman	Set by pump wavelengths (broad)	Limited by pump power (0.3 - 1 W)	Effective NF ≈ -1 to 0 dB	Pump-dependent, tunable with multi-pump	Pump-signal RIN transfer, low	Needs high pump power, safety (laser class 4), double-Rayleigh
Lumped Raman	Broad	As above	4 - 7 dB	Tunable	Low	Short, high-nonlinearity fibre; costly
Hybrid (Raman + EDFA)	Broad C-band	EDFA output +17 to +23 dBm	**Effective NF ≈ 0 dB (Raman) + EDFA**	Flattened by pump choice + GFF	Low	**Best OSNR; 5.5 dB better than EDFA-only here**

3.2 Analytical problem: output power and span gain

The question leaves the power levels blank, so the following assumed values are used: input signal per channel P_{in} = **-20 dBm** and target output per channel P_{out} = **0 dBm**, N = 32 channels.

Quantity	Equation	Result
Total output power (dBm)	$P_{tot,out} = P_{out} + 10 \log_{10} N = 0 + 10 \log_{10} 32$	**15.05 dBm**
Total output power (W)	$P = 10^{(15.05/10)}$ mW	**32.0 mW = 0.032 W**
Total input power	$P_{tot,in} = -20 + 15.05$	-4.95 dBm = 0.320 mW
Span gain required	$G = P_{out} - P_{in} = 0 - (-20)$	**20 dB** (a factor of 100)
Span loss it compensates	0.2 dB/km × 80 km + 4 dB connectors/splices	**20 dB** (so G = loss, unity net gain per span)

A 32 mW total output is easily delivered by a commercial C-band EDFA (saturated output +17 dBm = 50 mW or more), so output-power capability is not the constraint; the noise figure and the gain profile are.

3.3 Cumulative ASE noise and OSNR

For a chain of N identical spans the OSNR (0.1 nm) is $OSNR = 58 + P_{ch} - L_{span} - NF - 10\log_{10}N$, where $58\ dB = -10\log_{10}(h\nu B_{ref})$ at 1550 nm. For one EDFA-only span with 0 dBm launch, 20 dB loss and NF 5 dB:

$$OSNR_1 = 58 + 0 - 20 - 5 = \mathbf{33.0\ dB};\ \text{after } N = 8 \text{ spans: } 33.0 - 10\log_{10}8 = \mathbf{24.0\ dB}$$

For the Raman-assisted spans the signal power and the Raman ASE were integrated numerically along the 80 km fibre (equations $dP/dz = (g_R(z) - \alpha)P$ and $dN/dz = (g_R - \alpha)N + 2\,n_{sp}\,h\nu\,B\,g_R$, backward pump at 1450 nm, $C_R = 0.37\ W^{-1}km^{-1}$, $n_{sp} = 1.14$), then the 4 dB connector loss and the EDFA (NF 5 dB) that restores 0 dBm were added; 1.5 dB was subtracted for double-Rayleigh scattering and pump-noise transfer.

Configuration	Pump power	EDFA gain needed	OSNR, 1 span (dB)	OSNR, 8 spans (dB)	Margin over 17.8 dB (8 spans)
EDFA only (NF 5 dB)	-	20.0 dB	33.0	23.9	+6.1 dB
SOA only (NF 8.5 dB)	-	20.0 dB	29.5	20.4	+2.6 dB
Hybrid Raman 8 dB + EDFA	289 mW	12.0 dB	37.3	28.3	+10.4 dB
Hybrid Raman 10 dB + EDFA	362 mW	10.0 dB	38.4	29.4	+11.6 dB
Hybrid Raman 12 dB + EDFA	434 mW	8.0 dB	39.4	30.4	+12.5 dB

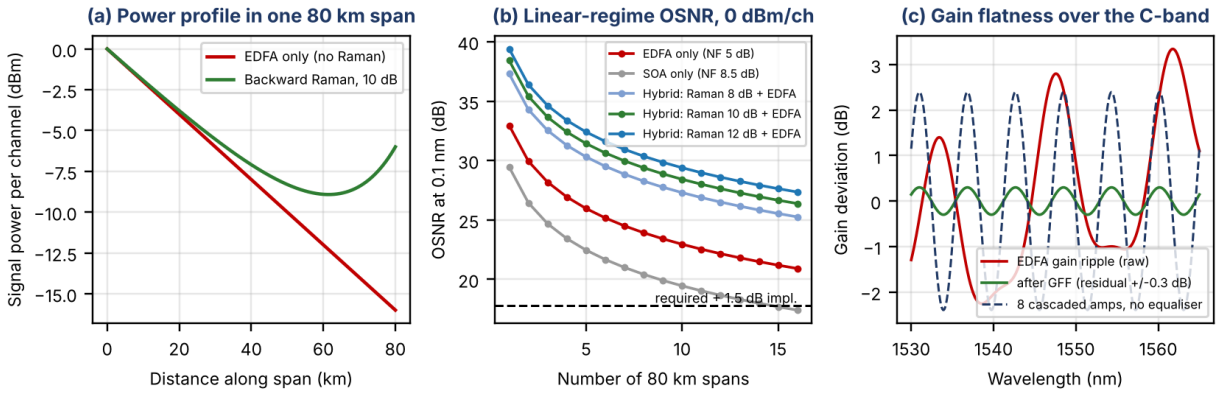


Fig. 3.1: (a) signal power along one span with and without backward Raman gain; (b) OSNR against the number of spans for each configuration; (c) gain ripple of the EDFA, the effect of the gain-flattening filter and the ripple of 8 cascaded amplifiers without equalisation

Interpretation. The Raman gain raises the signal inside the fibre (Fig. 3.1a: the minimum power is -9 dBm instead of -16 dBm), which is why its effective noise figure is close to 0 dB. A 10 dB Raman gain improves the OSNR per span by 5.5 dB over the EDFA alone (the 10 dB gain would give ~7 dB without the 1.5 dB allowance). Higher Raman gain (12 dB) adds only a further 1 dB but needs 434 mW of pump and increases double-Rayleigh and the average launch power; 8 - 10 dB is a good optimum. An SOA-only chain is 3.5 dB worse than the EDFA, and in addition suffers from fast gain dynamics that couple the 32 channels (cross-gain modulation and FWM), so it is rejected.

3.4 Gain non-flatness and the role of GFF / dynamic equalisation

The raw EDFA gain spectrum in the C-band has a peak-to-peak ripple of about 5.6 dB (Fig. 3.1c). A gain-flattening filter (a long-period fibre grating or thin-film filter) with the inverse profile reduces this to ±0.3 dB (0.6 dB peak-to-peak). But the residual errors of identical amplifiers add up almost coherently: after 8 amplifiers the worst-case excursion is 8 × 0.3 = **±2.4 dB** (4.8 dB peak-to-peak), so the weakest channel would lose 2.4 dB of OSNR while the strongest channel gets 2.4 dB more nonlinearity. The remedy is a **dynamic gain equaliser / wavelength-selective switch (WSS) every 4 spans** that re-flattens the 32 channel powers (reducing the excursion to ±1.2 dB) and a per-channel power control at the end; together with the Raman gain tilt tuning this keeps the channel-to-channel OSNR variation below 1 dB.

3.5 Recommended amplification architecture

DESIGN DECISION 2 - SELECT THE PROPOSED AMPLIFICATION CONFIGURATION

I recommend a hybrid amplifier: a backward-pumped distributed Raman stage (10 dB on-off gain, 1450 nm, about 360 mW) followed by a low-noise gain-flattened EDFA (NF 5 dB, 10 dB gain), with a dynamic gain equaliser every fourth span and a booster EDFA at the transmitter because it meets the power budget (20 dB span gain, +15.05 dBm total output = 32 mW, well within EDFA saturation power) and gives an OSNR of 38.4 dB per span, i.e. 29.4 dB after 8 spans, 11.6 dB above the 17.8 dB requirement, whereas the EDFA-only chain gives 23.9 dB (margin 6.1 dB, which, after nonlinearity, supports only 480 km with 2 dB margin - Stage 5). The SOA is rejected for its high noise figure and channel crosstalk; Raman-only operation needs impractical pump power and suffers double-Rayleigh noise. The GFF + equaliser limits the gain ripple to ±1.2 dB.

STAGE 4 - INVESTIGATE AND SELECT THE WDM IMPAIRMENT / DSP COMPENSATION CONFIGURATION

20 Marks | K4/K5 - Analyze and Evaluate

4.1 Physical mechanisms

Impairment	Mechanism and governing equation	Magnitude in this link
Chromatic dispersion	Group-velocity dispersion $\beta_2 = -D\lambda^2/2\pi c = -21.7\ ps^2/km$. Pulse broadening $\Delta\tau = D\ L\ \Delta\lambda$	$D\ L = 17 \times 640 = 10{,}880$ ps/nm; signal width $\Delta\lambda = \lambda^2 R_s/c = 0.23$ nm → $\Delta\tau = 2.51$ ns = **72 symbols** of ISI
Polarisation-mode dispersion	Random birefringence; mean DGD = $D_{PMD}\sqrt{L}$; Maxwellian statistics	$0.1 \times \sqrt{640} = 2.53$ ps = 0.07 symbol; 3× mean (outage) = 7.6 ps = 0.22 symbol
SPM	Nonlinear phase $\varphi_{NL} = \gamma P\ L_{eff}$, $L_{eff} = (1 - e^{-\alpha L})/\alpha$	$\gamma P\ L_{eff} = 1.3 \times 1$ mW $\times 25$ km $= 0.033$ rad per span (0.26 rad after 8 spans if added coherently)
XPM	Phase modulation by neighbouring channels' intensity: $\varphi_{XPM} = 2\gamma P_j L_{eff}$, reduced by walk-off	Strongest from the two nearest channels (50 GHz walk-off 17 ps/km)
FWM	New frequencies $f_{ijk} = f_i + f_j - f_k$; efficiency falls with D	Low: D = 17 ps/(nm km) phase-mismatches the products; included in the GN / SSFM results
PDL	Polarisation-dependent loss of isolators, couplers, AWG	~ 0.1 - 0.2 dB per component; allowed 0.5 dB margin

4.2 Optical-domain versus digital-domain compensation

Aspect	Dispersion-compensating fibre (DCF) / optical phase conjugation	Coherent receiver DSP
Principle	DCF: fibre with D ≈ -100 ps/(nm km) cancels accumulated CD ($L_{DCF} \approx 0.17$ L). OPC: mid-span spectral inversion reverses dispersion and nonlinearity	Linear equaliser with transfer function $\exp(-j\beta_2\omega^2 L/2)$
Added loss / noise	DCF: 0.5 - 0.6 dB/km → 6 - 10 dB per span, needs an extra amplifier (more ASE); higher nonlinearity (small A_{eff}). OPC: needs a pump, conversion efficiency, link symmetry	None (no extra optics)
Flexibility	Fixed to the fibre length and type; wavelength-dependent slope mismatch	Programmable: any length, any fibre, adapts to changes
PMD / polarisation	Cannot compensate (optical PMD compensators are narrow-band and complex)	Adaptive MIMO compensates PMD and polarisation rotation (up to hundreds of ps)
Nonlinearity	OPC compensates SPM only with exact symmetry	DBP possible, at heavy complexity
Cost / latency	Extra fibre spools, amplifiers; ~ 5 µs of extra fibre latency per 100 km	ASIC power; latency ~ 1 µs
Verdict	Not used - the uncompensated coherent link is better in OSNR, nonlinearity, cost and flexibility	**Adopted**

4.3 Operation of the key coherent DSP blocks

(1) Static chromatic-dispersion equalisation (FIR / frequency-domain)

Dispersion is an all-pass linear filter $H(\omega) = \exp(j\beta_2\omega^2 L/2)$; the receiver applies its inverse. The number of FIR taps for a sampling period T (2 samples/symbol, $T = 1/(2R_s) = 0.0$ ps) is $N = 2 \quad |D|L\lambda^2/(2cT^2) \quad + 1$ (Savory, 2008), giving **37 taps for 80 km, 145 for 320 km, 289 for 640 km** and 901 for 2000 km. A time-domain FIR filter therefore needs 578 complex multiplications per symbol per polarisation; an overlap-save **frequency-domain equaliser** (1024-point FFT) needs only about 31 - an **19× saving** - so FDE is chosen. The filter is static (it depends only on D and L), so it is placed first, before the adaptive blocks, which would otherwise have to track a very long impulse response.

(2) Adaptive polarisation demultiplexing and PMD compensation (MIMO)

After CD removal the two received polarisation tributaries are a linear mixture of the transmitted ones, $[E_x' \; E_y']^T = H(\omega)[E_x \; E_y]^T$ with the unitary Jones matrix H of the fibre (including PMD). A 2×2 butterfly of four complex FIR filters (h_{xx}, h_{xy}, h_{yx}, h_{yy}) estimates the inverse H^{-1}: $y_x = h_{xx}*x + h_{xy}*y$, $y_y = h_{yx}*x + h_{yy}*y$. The taps are updated blindly by the **constant-modulus algorithm** (Godard, 1980): $e = (R^2 - |y|^2)y$, $h \leftarrow h + \mu\, e\, x^*$. CMA converges without training because it needs only the knowledge that the amplitude should be constant; for 16-QAM, which has three amplitude rings, CMA is used for pre-convergence and then a radius-directed equaliser (RDE) with the three radii √0.2, 1 and √1.8 takes over. The mean DGD of 2.5 ps (maximum 7.6 ps) needs only 3 taps; 15 taps are used to allow for front-end skew and residual filtering (60 complex multiplications per symbol).

(3) Digital back-propagation (DBP) for nonlinear compensation

DBP solves the nonlinear Schrödinger equation backwards with the split-step Fourier method: for each step, a dispersive operator $\exp(-j\beta_2\omega^2 h/2)$ is followed by a nonlinear phase rotation $\exp(-j(8/9)\gamma|E|^2 L_{eff})$ (Manakov form). Complexity: per step, two FFT/IFFT, one dispersion multiplication and the nonlinear phase = about 13 complex multiplications per sample per polarisation; for 1 step per span over 8 spans at 2 samples/symbol this is about 208 multiplications per symbol (**6.8× the entire CD equaliser**), 416 for 2 steps/span (13.6×) and 1040 for 5 steps/span (34×). Single-channel DBP can remove only SPM; the XPM and FWM created by the other 31 channels are stochastic from the receiver's viewpoint, so the gain is limited. Full-field DBP would need all 32 channels to be received coherently at the same node.

The split-step simulation of this link (3 WDM channels, 8 × 80 km) measured the actual gain:

Launch power / channel	-2 dBm	0 dBm	+2 dBm	+4 dBm
Q (dB), MIMO only	8.31	8.22	7.47	5.97
Q (dB), + single-channel DBP (10 km steps)	8.49	8.57	8.49	7.89
Gain from single-channel DBP	+0.17 dB	+0.35 dB	+1.02 dB	+1.92 dB
Q (dB), + full-field DBP (3 channels, 2 km steps)	8.46	8.64	8.71	8.53

At the optimum launch power the single-channel DBP gain is only about 0.4 dB in Q (about 0.35 dB), while full-field DBP adds 0.4 dB. DBP mainly allows the launch power to be raised by 2 - 4 dB (Q stays above 8 dB at +2 dBm), which pays off only on very long routes; at 7× the DSP complexity it is not justified for 640 km. (Single random seed per point: differences of ±0.15 dB are within simulation noise.)

4.4 Recommended coherent receiver pipeline

Fig. B: Recommended coherent DSP receiver pipeline

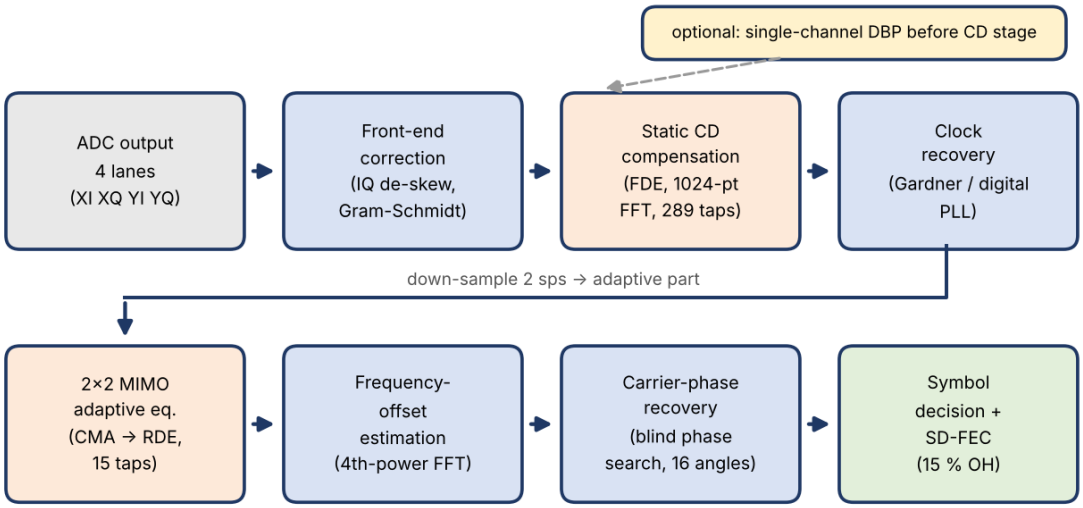


Fig. 4.1: Coherent DSP receiver chain recommended for the 200G DP-16QAM channel

#	Block	Reason for its position in the sequence
1	ADC output → front-end correction (IQ skew, orthogonalisation)	Hardware imperfections of the hybrid / PD / ADC must be removed before any algorithm assumes ideal IQ signals
2	Static CD compensation (FDE)	Long, fixed, signal-independent filter; has to be applied before the polarisation filter, because CD acts identically on both polarisations and is largest (≈ 290 taps) - the MIMO filter would otherwise need hundreds of adaptive taps
3	Timing (clock) recovery	Needs the CD-free eye to find the sampling phase; decimation to the processing rate
4	2×2 MIMO equaliser (CMA → RDE)	Separates polarisations and removes PMD, residual CD and filtering ISI; works at 2 sps and is independent of carrier phase (the CMA error is phase-insensitive), hence before carrier recovery
5	Frequency-offset estimation	Laser detuning (up to ± 1 - 2 GHz) is removed first so that phase estimation does not slip
6	Carrier-phase estimation (blind phase search, 16 test angles)	Removes laser phase noise (Δν = 200 kHz) that rotates the constellation; needs a stable amplitude and frequency, so it comes last
7	Decision + SD-FEC (15 % overhead)	Corrects the residual BER up to 2×10^{-2} to below 10^{-15}

DESIGN DECISION 3 - SELECT THE DSP COMPENSATION CONFIGURATION

I recommend a fully digital coherent receiver (no DCF, no optical PMD compensator): frequency-domain static CD equalisation (1024-point FFT, 289 taps equivalent), 2×2 MIMO CMA/RDE equaliser with 15 taps for polarisation demultiplexing and PMD, frequency-offset and blind-phase-search carrier recovery, and 15 % SD-FEC; single-channel DBP is reserved as an optional firmware upgrade because it removes 72 symbols of CD spread and 2.5 ps of PMD with 31 + 60 complex multiplications per symbol, adds no loss or noise, adapts automatically, and in simulation recovers a clean 16-QAM constellation at 640 km (Q = 8.2 dB, BER 5.0e-03, compared with the 6.25 dB threshold). DBP adds only about +0.35 dB Q at the optimum power (+1.0 dB at +2 dBm) for roughly 7 times the CD-block complexity, so it is not adopted in the baseline.

STAGE 5 - EVALUATE THE INTEGRATED WDM TRANSMISSION STRATEGY

15 Marks | K5 - Evaluate

5.1 Integrated system and global performance parameters

Fig. A: Block-level architecture of the 32-channel, 6.4 Tb/s DWDM link

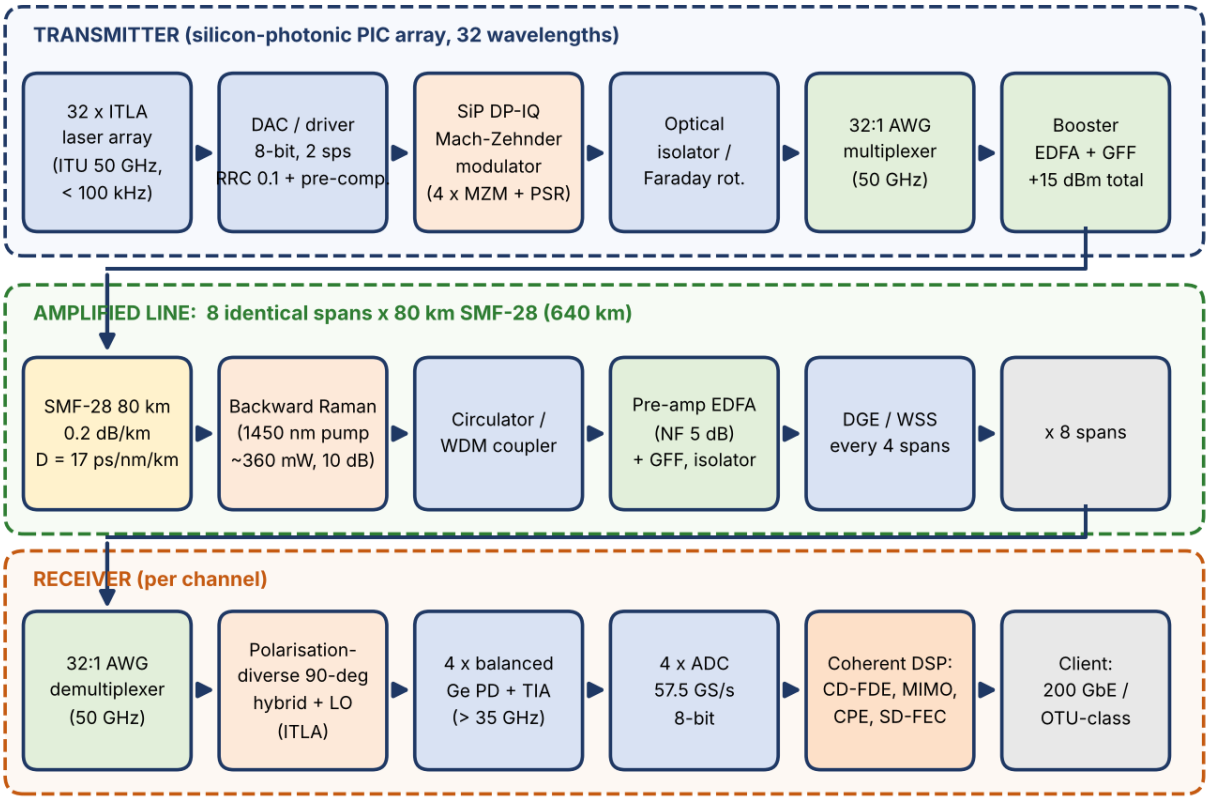


Fig. 5.1: Integrated transmitter - amplified line - receiver architecture

Global parameter	Value	Basis
Total aggregate throughput	6.4 Tb/s net (7.36 Tb/s line)	32 × 200 Gb/s
Net spectral efficiency	4.0 b/s/Hz	200 Gb/s / 50 GHz
System Q-factor (640 km, 0 dBm)	8.67 dB (GN model); 8.22 dB (split-step simulation)	Fig. 6.1; BER 3.3e-03 / 5.0e-03
OSNR margin over FEC threshold	29.4 dB available (ASE only) vs 16.3 dB required → 13.1 dB; after nonlinearity and transceiver noise, SNR margin = 2.6 dB	SNR budget: ASE 25.8 dB, NLI 23.8 dB, transceiver 16.5 dB → 15.3 dB vs 12.7 dB required
Power consumption per bit (estimate)	~ 110 pJ/bit per transceiver (22 W / 200 Gb/s); ~ 44 pJ/bit for the 8-span line (~ 280 W / 6.4 Tb/s); ~ 0.27 W/Gb/s for the whole bidirectional link	Typical industry figures for a 200G coherent CFP2-DCO class module, EDFA + Raman pump drivers, WSS
Footprint (estimate)	Tx PIC die ~ 6 × 4 mm; 32 CFP2-DCO-class modules fit in ~ 2 - 4 rack units; amplifier site ~ 1 - 2 RU	Form-factor data (CFP2 = 41.5 × 107.5 × 12.4 mm)

5.2 Inter-dependencies

How the modulation order dictates amplifier OSNR and DSP complexity. At constant net rate, raising the modulation order from QPSK to 16-QAM to 64-QAM raises the required OSNR from 12.9 to 16.3 to 20.3 dB, i.e. every extra 2 bits per symbol cost about 3.5 - 4 dB, which must be paid by lower amplifier noise (Raman instead of EDFA), shorter spans or a shorter route. The same step reduces the baud rate from 57.5 to 28.75 to 19.2 GBd, which reduces the CD filter length (the tap count is proportional to R_s^2: 289 taps at 28.75 GBd versus about 1156 at 57.5 GBd) and the ADC speed, but increases the sensitivity to phase noise (16-QAM allows only about 0.1 rad rms), IQ imbalance, transceiver SNR (needs 12.7 dB for 16-QAM, 18.4 dB for 64-QAM) and nonlinearity, and requires a more complex carrier-phase (BPS) and equaliser (RDE rather than plain CMA) stage.

How MMI coupler losses impact launch power. The transmitter PIC has about six 1×2 / 2×1 MMI stages in the optical path (polarisation split, IQ split, arm splits and combiners); with 0.3 dB excess loss each this alone is 6 × 0.3 = 1.8 dB. The path budget is: laser +14 dBm − coupling 1.5 dB − polarisation split 3 dB − IQ combination 3 dB − MMI excess 1.8 dB − modulation loss 6 dB − waveguides 1 dB − output coupler 1.5 dB = **-3.8 dBm per channel**. The 32:1 AWG (4.5 dB) leaves -8.3 dBm per channel, so the booster must provide 8.3 dB of gain to reach the 0 dBm launch power. Every dB of coupler loss is replaced by 1 dB more booster gain; this costs OSNR only if the signal reaching the booster is too weak: its input OSNR is 58 + (-8.3) - 5 = 44.7 dB, far above the link OSNR of 29.4 dB, and the combined end-to-end OSNR is 29.3 dB. If the 32:1 multiplexer were built as a binary MMI tree (10 $\log_{10}$32 + 5 × 0.3 = 16.6 dB loss), the signal reaching the booster would be -20.4 dBm, the transmitter OSNR drops to 32.6 dB, and the end-to-end OSNR falls to 27.7 dB, a **1.6 dB penalty**. Hence the AWG is used for multiplexing, and MMIs only inside the modulator.

5.3 Trade-offs between reach, channel density, dynamic range and feasibility

Number of spans	Distance	Q (dB) hybrid, optimum power	Margin over 6.25 dB	Q (dB) EDFA only, optimum power	Margin
4	320 km	9.26	+3.0 dB	8.80	+2.5 dB
6	480 km	9.03	+2.8 dB	8.39	+2.1 dB
8	640 km	8.81	+2.6 dB	8.02	+1.8 dB
10	800 km	8.61	+2.4 dB	7.68	+1.4 dB
13	1040 km	8.31	+2.1 dB	7.21	+1.0 dB
16	1280 km	8.04	+1.8 dB	6.80	+0.6 dB

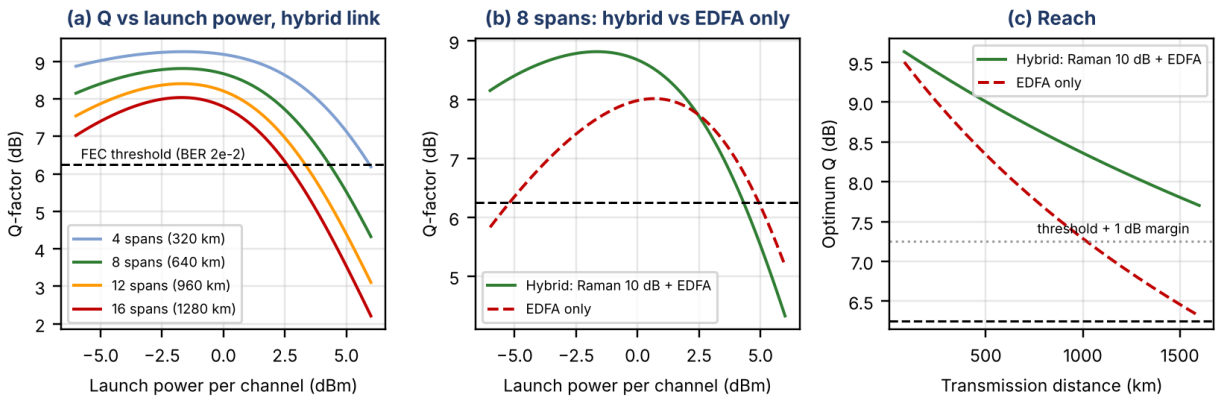


Fig. 5.2: (a) Q-factor against launch power for 4 to 16 spans; (b) hybrid versus EDFA-only amplification over 8 spans; (c) optimum Q against distance (GN model)

- **Reach.** With a 2 dB total margin the hybrid design reaches 1040 km (13 spans) versus 480 km for EDFA-only: the Raman stage doubles the reach. The 640 km design point is well inside this (analytical margin 2.6 dB at the optimum power).
- **Channel density.** Narrowing the spacing to 37.5 GHz would give 5.3 b/s/Hz and 40 channels in 1.5 THz, but leaves only 5.9 GHz guard band (RRC roll-off 0.1 at 28.75 GBd), so a flex-grid ROADM filter and an extra ~ 0.5 dB crosstalk penalty would be required. Extending to the full C-band (88 channels at 50 GHz) keeps the same reach but needs 2.75× more pump and booster power.
- **Dynamic range.** The optimum Q is flat to about ±2 dB of launch power (Fig. 5.2a/b), so power drift, channel add/drop and tilt can be absorbed; beyond +4 dBm per channel nonlinearity drops the Q below the threshold (simulation: Q = 6.0 dB at +4 dBm).
- **Deployment feasibility.** All building blocks (CFP2-DCO-class transceivers, EDFA with GFF, Raman units) are commercially available technologies; the only non-standard element is the use of a 28.75 GBd line rate (commercial 200G modules use 56 - 60 GBd DP-QPSK/8QAM/16QAM); a custom 28.75 GBd 16QAM is feasible with the same DSP technology.

5.4 Critical failure points, scalability limits and compliance

Item	Risk / limit	Mitigation
Tx laser wavelength drift	> ± 2.5 GHz causes inter-channel crosstalk at the 50 GHz AWG and WSS	Wavelength locker, thermal control, ± 1.25 GHz specification
Raman pump failure / back-reflection	Loss of 10 dB gain in the span; high optical power (laser safety)	Redundant pump, automatic laser shutdown (ALS), EDFA gain re-adjusts to hold 0 dBm for at least 1 - 2 spans
Gain ripple accumulation	± 2.4 dB per 8 amplifiers	WSS equaliser every 4 spans
Modulator drive and bias drift	$V_{\pi} \approx 4.5$ V, bias drift of the MZM, IQ imbalance	SiGe driver 3 V_{pp}, closed-loop bias control, receiver IQ-imbalance compensation
Nonlinearity at high power or low D (new fibre types)	NLI grows as P^3 and with dispersion-shifted fibre	Power-optimisation control, DBP upgrade, G.655 fibres checked case by case
Scalability limit	About 88 channels in C-band; 6.4 Tb/s → 17.6 Tb/s	Add L-band and a second Raman pump set
Single point of failure	Mux / demux AWG, booster EDFA	Passive AWG (high MTBF); 1+1 amplifier protection

Standards compliance. (i) **ITU-T G.694.1**: all 32 carriers lie on the 50 GHz DWDM grid anchored at 193.1 THz; (ii) **ITU-T G.698.2** (amplified multichannel DWDM interfaces): the design follows the same application model (transmitter, optical-amplified link, receiver with specified OSNR and BER limits); (iii) **IEEE 802.3ck / OIF-CEI-112G**: the 200 GbE host side is 2 × 100 Gb/s (106.25 GBd PAM4) electrical lanes, which the line-card gearbox / DSP interface connects to the optical module; (iv) **IEEE 802.3ct and OIF 400ZR** define coherent DWDM line-side PHYs (100G, 400G); the 200G, 28.75 GBd DP-16QAM format used here is not a defined IEEE PHY, so line-side interoperability is **proprietary** (single-vendor), although the 15 % SD-FEC, framing and grid follow the OIF approach. This is the main compliance limitation and is stated honestly in the final decision.

DESIGN DECISION 4 - CREATE THE INTEGRATED WDM STRATEGY

I recommend adopting the integrated 32-channel DP-16QAM, silicon-photonic-transmitter, hybrid Raman/EDFA, coherent-DSP architecture for deployment on routes up to 640 km (8 × 80 km) because the cross-stage evidence is consistent: the format (Stage 2) needs 16.3 dB OSNR; the amplifiers (Stage 3) deliver 29.4 dB; the DSP (Stage 4) removes CD and PMD; and the end-to-end SNR of 15.3 dB (analytical) / 14.8 dB (simulation) exceeds the 12.7 dB threshold by 2.1 - 2.6 dB. Throughput is 6.4 Tb/s at 4 b/s/Hz and about 0.27 W/Gb/s, and the Raman stage doubles the reach compared with EDFA-only. The 1.5 dB end-of-life allowance leaves a spare of 0.5 - 1.1 dB at 640 km, which is why the deployment is limited to 640 km.

STAGE 6 - VERIFY AND CRITICALLY EVALUATE THE COMPLETE WDM SYSTEM

10 Marks | K5/K6 - Evaluate and Create/Design

6.1 Verification table

Requirement / performance parameter	Specified / target value	Achieved / evidence value	Status
Aggregate WDM capacity	6.4 Tb/s (32 ch × 200 Gb/s)	6.4 Tb/s net; 7.36 Tb/s line (32 × 230 Gb/s)	Met
Spectral efficiency and modulation choice	DP-16QAM, 4 b/s/Hz on 50 GHz	DP-16QAM at 28.75 GBd, 4.0 b/s/Hz net (4.6 gross); occupied bandwidth 31.6 GHz	Met
Modulator bandwidth and footprint	≥ 20 GHz; compact PIC	35 GHz (SiP pn MZM); 4 mm per MZM, PIC die ~ 6 × 4 mm; V_{π} 4.5 V; eye opening 32 % at 30 GHz limit (PAM4 at 115 GBd: 2 %)	Met (drive voltage is the weak point)
Amplifier gain and gain flatness	G = 20 dB per span, 0 dBm/ch, ripple < ± 1.5 dB	Raman 10 dB + EDFA 10 dB = 20 dB; +15.05 dBm total; ripple ± 0.3 dB per amplifier, ± 1.2 dB with WSS every 4 spans	Met
OSNR	≥ 17.8 dB (0.1 nm) incl. 1.5 dB implementation	29.4 dB after 8 spans (EDFA-only: 23.9 dB)	Met (+11.6 dB)
Pre-FEC BER / Q-factor	BER ≤ 2×10^{-2} (Q ≥ 6.25 dB)	BER 3.3e-03, Q = 8.67 dB (GN); BER 5.0e-03, Q = 8.22 dB (simulation, back-to-back 9.09 dB)	Met (+2.4 / +2.0 dB)
DSP impairment mitigation	Full CD (10,880 ps/nm), PMD (> 7 ps) and nonlinear tolerance	CD: 289-tap equivalent FDE; PMD: 15-tap MIMO; nonlinear: optimum power plus optional DBP (+0.35 dB at optimum, +1.0 dB at +2 dBm)	Met
Standards and feasibility	ITU-T G.698 and IEEE 802.3ck compliance	50 GHz grid (G.694.1/G.698.2) and 802.3ck host interface compliant; line side is proprietary (no 200G DP-16QAM PHY in 802.3ct)	Partly met

6.2 Simulation results

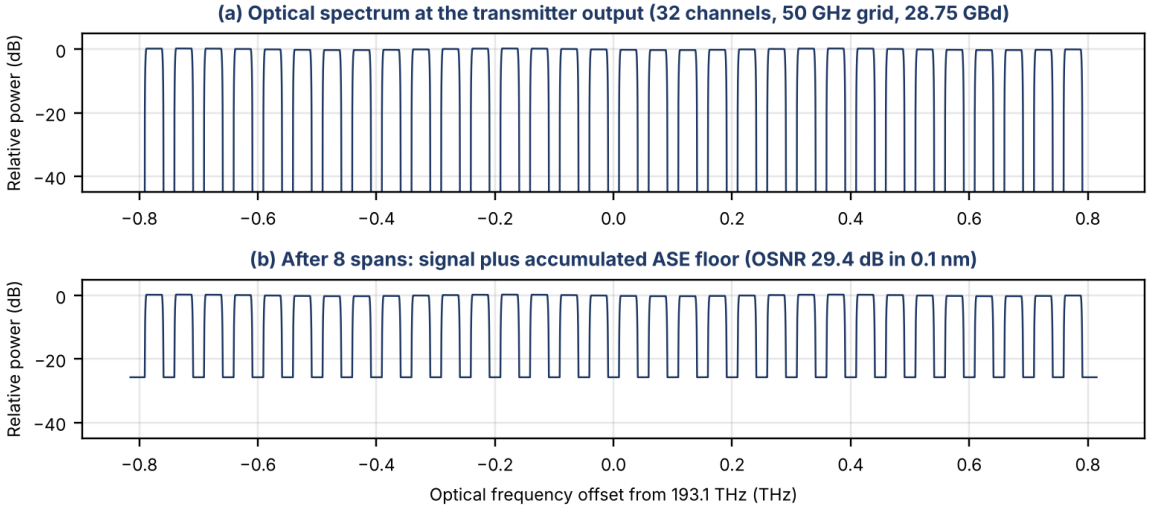


Fig. 6.1: Optical spectra of the 32-channel WDM signal at the transmitter output and after 8 spans (ASE floor corresponding to the computed OSNR)

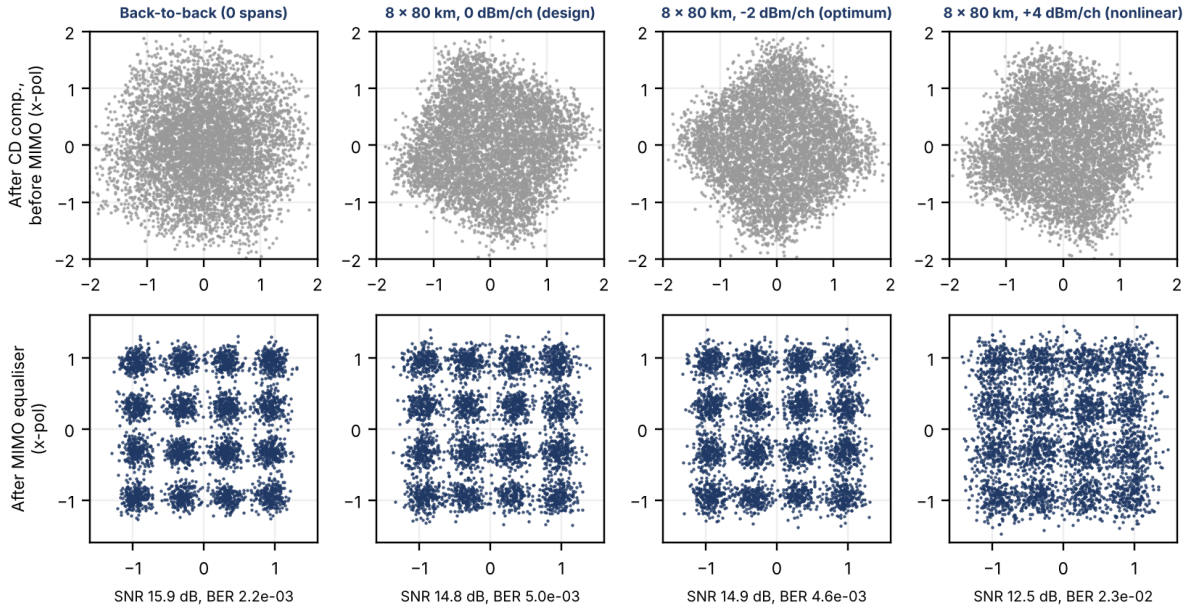


Fig. 6.2: Constellations (x polarisation, centre channel) before and after the MIMO equaliser: back-to-back and after 8 × 80 km at three launch powers

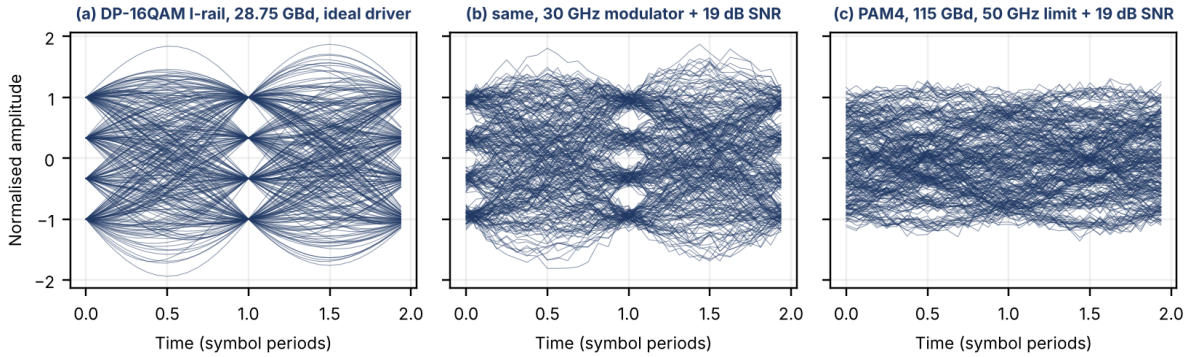


Fig. 6.3: Eye diagrams of the in-phase rail: (a) ideal DP-16QAM driver; (b) with 30 GHz modulator bandwidth and 19 dB SNR; (c) PAM4 at 115 GBd limited to 50 GHz - the 4-level eye is closed

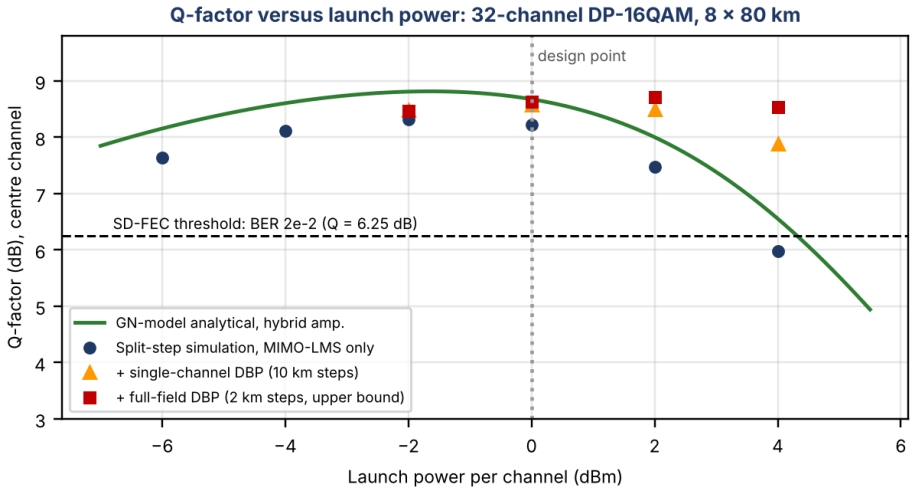


Fig. 6.4: Q-factor versus launch power - analytical GN model and split-step simulation (centre channel of a 3-channel WDM), with and without digital back-propagation

Simulation case (centre channel, DP-16QAM)	SNR (dB)	Pre-FEC BER	Q (dB)
Back-to-back (transceiver only)	15.85	2.2e-03	9.09
8 × 80 km, -2 dBm/ch (near optimum)	14.91	4.6e-03	8.31
8 × 80 km, 0 dBm/ch (design)	**14.77**	**5.0e-03**	**8.22**
8 × 80 km, +4 dBm/ch (nonlinear regime)	12.47	2.3e-02	5.97

The simulation is 0.5 dB below the analytical GN prediction at 0 dBm: the 3-channel simulation contains less XPM than a 32-channel system would, while the finite-length MIMO equaliser and LMS training add a small penalty, so the GN model (which assumes a full 32-channel load) and the simulation are consistent to within about 0.5 dB. The conclusion of the verification is the lower, simulated margin of 2.1 dB (SNR) / 2.0 dB (Q). **Limitations of the simulation:** only 3 of the 32 channels were propagated; laser phase noise and ADC quantisation were not included (carrier-phase estimation is treated as ideal); the equaliser was trained on 16,384 known symbols and tested on the next 15,000 (a pilot-aided stand-in for the blind CMA/RDE of Stage 4); the Raman amplifier was represented by its measured power profile and a lumped ASE source; each point is a single random realisation of 32,768 symbols per polarisation (≈ 400 - 1,000 errors per BER point).

6.3 Final system design

Fig. D: Simulation layout (OptiSystem / OptSim style) - component-level schematic

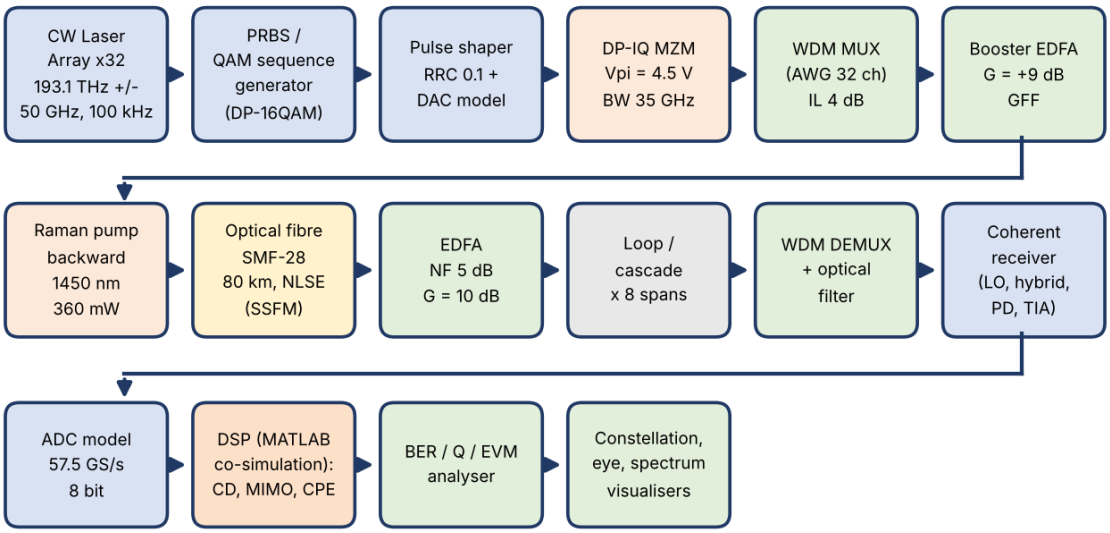


Fig. D: OptiSystem / OptSim-style component-level layout of the simulated link

Fig. C: Silicon-photonic dual-polarisation IQ transmitter (one wavelength)

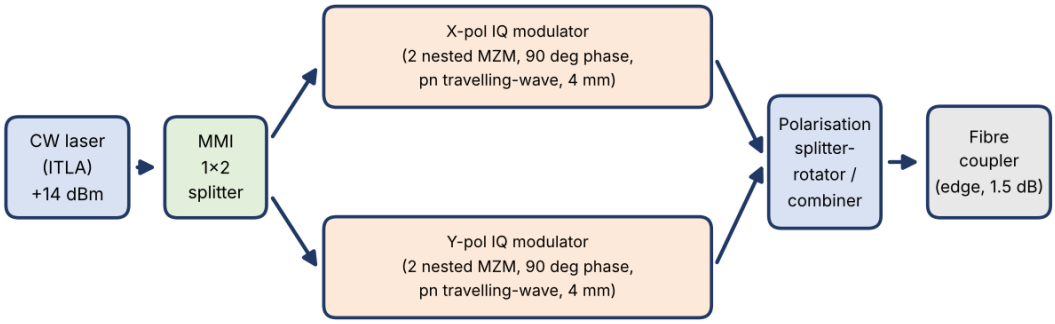


Fig. C: Silicon-photonic dual-polarisation IQ transmitter of one wavelength

Complete system design parameter table

Subsystem	Parameter	Value
WDM	Channels / grid / carrier range	32 / 50 GHz (ITU-T G.694.1) / 192.325 - 193.875 THz (1546.3 - 1558.8 nm)
Laser	Type / linewidth / output power / RIN	External-cavity ITLA / < 100 kHz (Tx) + < 100 kHz (LO) / +14 dBm / < -145 dB/Hz
Modulation	Format / baud / line rate / roll-off	DP-16QAM / 28.75 GBd / 230 Gb/s / RRC $\beta = 0.1$
Modulator	Type / V_π / bandwidth / insertion loss	SiP DP-IQ MZM, pn carrier depletion / 4.5 V ($V_\pi L$ 1.8 V cm, L = 4 mm) / 35 GHz / ~ 12 dB total Tx PIC loss
Driver / DAC	Swing / resolution / sampling	3 V_{pp} per arm (SiGe) / 8 bit / 57.5 GS/s (2 sps)
Mux / demux	Type / loss / channel width	32-channel AWG (silica or SiN) / 4.5 dB / 0.5 dB bandwidth > 35 GHz
Booster EDFA	Gain / output / NF	8.3 dB / +15 dBm total (0 dBm per channel) / 5 dB
Fibre	Type / span length / spans	SMF-28: $\alpha = 0.2$ dB/km, D = 17 ps/(nm km), S = 0.056 ps/(nm^2 km), $\gamma = 1.3$ W^{-1}km^{-1}, $A_{eff} = 80$ µm^2, PMD = 0.1 ps/√km / 80 km / 8
Raman	Pump / gain / effective NF	1450 nm backward, ~ 360 mW / 10 dB on-off / ~ -1 dB
In-line EDFA	Gain / NF / GFF	10 dB / 5 dB / residual ripple ± 0.3 dB; WSS equaliser every 4 spans
Receiver	Hybrid / PD / TIA	Polarisation-diverse 90° hybrid, Ge PIN > 35 GHz, responsivity 0.9 A/W, TIA 40 GHz
ADC	Rate / resolution / ENOB	57.5 GS/s / 8 bit / > 5.5 bits
DSP filters	Taps	CD: 289 (FDE, 1024-pt FFT); MIMO: 15 taps × 4; BPS: 16 test angles, window 21; SD-FEC 15 % OH
Launch condition	Power per channel / total	0 dBm / +15.05 dBm (nonlinear optimum -2 to 0 dBm)
Targets	OSNR / Q / FEC limit	29.4 dB / 8.7 dB / BER 2×10^{-2}

Final engineering decision

WOULD YOU DEPLOY THE INVESTIGATED AND DESIGNED WDM SYSTEM? YES - WITH TECHNICAL JUSTIFICATION AND CONDITIONS.

- **Modulation efficiency and OSNR sensitivity:** DP-16QAM carries 200 Gb/s on 28.75 GBd and 50 GHz (4 b/s/Hz). It needs 16.3 dB OSNR (SNR 12.7 dB) - 3.5 dB above QPSK - but the link supplies 29.4 dB, so the higher efficiency is 'free' at 640 km, and PAM4/OOK are physically impossible within 50 GHz.
- **Photonic modulator integration, bandwidth and drive voltage:** a silicon DP-IQ MZM provides 35 GHz (> 20 GHz needed), is chirp-free and CMOS-integrable; the weakness is $V_\pi \approx 4.5$ V and ~ 12 dB total insertion loss, which requires a 3 V_{pp} driver and 9 dB of booster gain - both achievable. Micro-rings were rejected (0.08 nm/K drift against a 0.19 nm linewidth).

- **Amplification, noise accumulation and power budget:** a 20 dB span gain and 32 mW total output are comfortably within EDFA capability; the hybrid Raman-EDFA stage lifts the 8-span OSNR from 23.9 dB to 29.4 dB and doubles the reach at 2 dB margin (480 → 1040 km). Gain ripple is held to ± 1.2 dB with GFFs and a WSS every 4 spans.
- **DSP effectiveness:** FDE removes 72 symbols of dispersion, the MIMO equaliser handles PMD and polarisation, and the simulated constellation after 640 km is clean (Q 8.2 dB, BER 5.0e-03 < 2×10^{-2}). Nonlinearity limits the launch power to about +2 dBm per channel; single-channel DBP is optional because it gives only ~ 0.35 dB at the optimum power.
- **Power, scalability and cost:** about 0.27 W/Gb/s for the bidirectional link (110 pJ/bit per transceiver), mostly in the DSP ASIC; the 1.6 THz used leaves 2.8 THz for future channels, and silicon photonics keeps the transceiver cost per bit low in volume.

Conditions of deployment. (1) Limit routes to **8 × 80 km = 640 km**: the measured SNR margin (2.1 dB simulated, 2.6 dB analytical) must cover the 1.5 dB end-of-life allowance (0.5 dB ageing and repairs, 0.5 dB PDL and filtering, 0.5 dB temperature and tolerances). (2) Commission at -1 to 0 dBm per channel with automatic power optimisation. (3) The line-side interface is **not** an IEEE-standardised PHY: use one vendor family on both ends or standardise on OIF 400ZR-style 16-QAM at 59.8 GBd if multi-vendor interoperability is required. (4) Before final purchase repeat the verification in OptiSystem/OptSim with all 32 channels and measured component data.

If the margin is judged insufficient (for example, an operator who demands 3 dB EOL margin), the redesign path is: (i) shorten to 6 spans (480 km): optimum Q rises to 9.03 dB (margin +2.8 dB); or (ii) raise the Raman gain to 12 dB (+1.0 dB OSNR per span); or (iii) switch on single-channel DBP and launch at +2 dBm (+1.0 dB Q, simulated); or (iv) lower the order to DP-8QAM / DP-QPSK at 57.5 GBd on a 75 GHz grid (6.4 Tb/s → 4.3 Tb/s, with 3.5 dB lower OSNR requirement) to reach over 2000 km.

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CODE OF CONDUCT - I **AJAY G** (Name) / **192412161** (Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____